

Dev log

22nd May

Created a basic tile system, the game will attempt to fill the user's screen by a 40x40 grid of tiles, at which point it will then pick two points, a start and end point. The user will be able to make a path from one point to the and it will be saved into an array that can store each path in sequence. Made a basic movement system that uses WASD to move around

BUG: Game crashes if user makes path that goes out screen, will fix tomorrow with a quick check

26th May

Worked on the actual path making system, at this time a separate class called pathmaker holds the position of the user's current tile, TODO: merge into scene game as it holds so little data it could probably be removed entirely. Added new functions to pathlist to check if the player has reached the end of a given path yet. Added new functions to the tile class to allow it to see which was its previous tile and next tile so that it can help enemies keep track of where to go next

28th May

Realized I had a bug in my framework where I updated the keypresses twice instead of once, causing inputs that should've counted as "pressed" to instead be counted as held. Also added box2d functionality to the game so that we can check for collisions between enemies, projectiles, towers etc.

29th May

Added the tower class. At this time it's a basic ball sprite with regular stats (firerate, damage, speed etc.), however I've coded it so that its information is modular and can be switched out later down the line. This tower can keep track of enemies (turns them red when in range) and when they leave the radius (turns blue to indicated left radius) (will remove later this is just for debugging)

30th May

Polished up the tower class and merged it into main. Need to now plan how I want projectiles to work

31st May

Had to fix the tower class, as I forgot to make sensor events check if that event was relevant to a given tower, causing all towers to think an enemy leaving 1 tower's radius meant it left every other tower's radius as well. Moved text to be initialized later as it still corrupts sprites if initialized first (to look into)

1st June

Set up a basic projectile class that's a sensor and will damage the enemy when close enough. It receives the enemy its targeting as an input from the parent (the tower) upon initialization, and if not homing travels to the enemy's position where it was first seen. If homing, it'll keep track of

this enemy until it hits it or flies in a straight line if the enemy is deleted mid homing. At this time, the stats are hard coded, and need to be typed in manually, I'll add an INI parsing system so we can just easily slot in projectiles/ towers by passing in an ID for it to reference.

2nd June

Added an INI Parser that allows the user to add a given projectile/tower/enemies through a basic format. This format goes as follows (tower for example):

[Shooter]

Sprite=Shooter

Price=10

Range=1

Firerate=2

ProjectileID=Dart

Speed=1

Also I figured out what the texture problem with text was, as it turns out the formatting of the text sprite never got switched off so it attempted to load all textures as if it was loading text instead of the regular format. I also added turning to the projectiles and towers so theyd face the direction of the enemy they're aiming at for easier clarity. I also added effects to the projectiles class so that enemies hit by a certain type of projectile will behave differently (damage over time, slowed down movement)

I also made it so homing projectiles will attempt to look for another target in the owner's range if its dead. I don't want to make them look for enemies in its own range as it would mean that spamming homing towers all over the map instead of anywhere near enemies would be a viable strategy.

4^h June

Bug fixes (Enemies were on 2x speed and homing needed work)

5th June

Added boomerang tower that pierces infinitely and shoots a projectile in a straight line before flying back

Added a detonator tower that does big damage in a small radius with a long cooldown

Added new sprites so the start and end tile is a unique flag+hole sprite instead of just being blue and red. Also added a cursor so the player can tell where they are easier (adding this improved UI convenience a ton)

Fixed towers waiting for their fire delay time upon being placed instead of just having their cooldown start at 0 and THEN wait after firing the first time

Adjusted the prices because some powerful towers were too cheap

7th June

Bug fixes for enemy collision

Kinematic bodies work differently to dynamic bodies in that they don't automatically move other bodies and can only be moved manually, which is what we were doing in the game so now enemies shouldn't block those behind them

8th June

Added new sprites to add a bit more polish to the game, added new digging noises that are randomly picked to add variance. Added random chance for tiles to be obstacles so that the user cant just go in a straight line each time. This cant be placed or pathed on.